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Introduction of a panel component in the LUCAS survey and the use of a composed estimator for statistics production

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MANUALS AND
GUIDELINES

Introduction of a panel component in the LUCAS survey and the use of a composed estimator for statistics production

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Ballin Marco, Barcaroli Guilio, Palmieri Alessandra, Zachariadis Savvas

The paper presents a statistical study aimed at determining the optimal rate of panel sample points for the European Land Cover and Land Use Survey (LUCAS) survey. Furthermore, the pros and cons of adopting a composed estimator for statistics production by taking full advantage of a panel component are investigated

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Abstract

Each of the LUCAS samples, from the first round (2009) to the most recent in 2022, was designed as a cross-sectional sample of the entire frame (the Master Sample). Each round featured a subset of points shared with the previous round. The maximum overlap rate occurred in 2015 (87 %), while the minimum was in 2018 (34 %). Intermediate rates were 63 % in 2012 and 40 % in 2022. The incidence of overlapping components was either unplanned or, if planned, not based on a rigorous methodology to determine the optimal rate for maximising estimate accuracy. This paper presents a study aimed at determining this optimal rate. Two types of estimates will be considered:

(a) level estimates, which concern the totals or means of parameters of interest at a specific date (e.g., the percentage of artificial land cover in 2022); and

(b) change estimates, which describe variations in the totals or means of parameters of interest between two dates (e.g., the increase or decrease in the percentage of artificial land cover from 2018 to 2022).

Furthermore, the pros and cons of adopting a composed estimator that takes full advantage of the presence of a panel component will be investigated. The first issue (optimal panel component rate) is analysed using a simulation study. In contrast, the second issue (the convenience of adopting the composed estimator) is based on comparing the accuracy of the current calibration estimator with that of the composed estimator used in the LUCAS 2022 round estimates. In addition, the application of a specific function of the R package ReGenesees, namely the svyDelta function, which facilitates the production of change estimates with accurate calculation of the related sampling variance, is illustrated.

Finally, general guidance is provided regarding the potential introduction of a panel component in the LUCAS survey.

Keywords: land cover, land use, panel, level and change estimates, composed estimator.

JEL Classification: C8 (Survey Methods • Sampling Methods)

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1 Introduction

Within the context of improving quality and completeness of the Land Cover and Land Use statistics, Eurostat has conducted the LUCAS survey every three years since 2006, except for the most recent round (2022, held four years after the previous round, in 2018). The surveys are used to monitor the social and economic use of land as well as ecosystems and biodiversity. Sustainable development indicators and agro-environmental indicators for soil are examples of LUCAS data use, while the microdata collected also serve to produce, verify, and validate outputs for the Copernicus Programme.

The LUCAS survey is based on a sample of geo-referenced points, selected from a frame of more than 1 million points at the intersections of a 2 square km grid laid over the entire EU territory, known as the Master Sample or first-phase sample (Ballin et al., 2018). It is carried out by two data collection modes: direct observations by surveyors in a small area centred around the selected point (a circle with a radius of 1.5 meters or, in some cases, 20 metres), and photo interpretation. In the first case, for each point, information is collected on Land Cover (i.e., the bio-physical cover of the land, such as natural areas, forests, buildings, roads, and lakes), on Land Use (i.e., the socio-economic use of the land, such as agriculture, commerce, residential use, or recreation), and on other variables linked to Land Cover.

In addition to the LUCAS core data, each survey also collects specific information called ‘ad hoc modules’: this was the case for the topsoil sample in 2009, 2015, 2018, and 2022; the transects in 2009, 2012, and 2015; the grassland module; and the Copernicus supplementary points in 2018 and 2022. The topsoil sample in 2022 had a panel component with respect to the 2018 one (Ballin et al., 2022).

The LUCAS survey is a cross-sectional survey, meaning that at each round, a sample is selected from the frame that is formally independent of previously selected samples, and no panel component is planned. Nonetheless, important overlaps can be observed between consecutive samples, and the condition of complete independence is not entirely met, as many of the overlapping points are selected from take-all strata with certainty. This defines an implicit panel component between the two rounds.

In this paper, the convenience of introducing a planned panel component in the LUCAS survey is investigated, and a possible strategy for its introduction is proposed.

In Chapter 2, the methodology for determining the optimal panel component rate is introduced and applied using data collected in the 2018 and 2022 rounds of the survey. Chapter 3 describes the composed estimator, which leverages the presence of a panel component to increase the accuracy of both change estimates and level estimates: a comparison with the usual calibration estimator is provided. In Chapter 4, a method for calculating the sampling variance for change estimates in the presence of a panel component is illustrated and applied to the specific case of changes in artificial land cover areas from 2018 to 2022. Finally, Chapter 5 draws conclusions regarding the different possible alternatives for introducing a panel component in the LUCAS survey.

2

Determination of the panel component rate

A panel component in periodic surveys involves repeatedly surveying the same individuals or groups (panel) over time. This approach allows researchers to track changes, understand trends, and gain deeper insights into the dynamics of variables under study. Panel components can be integrated within broader cross-sectional surveys to combine the benefits of both methodologies (see Lynn, 2009, and Lipps, 2021).

Among the advantages of using a panel component in a periodic survey are the following:

1. *Trend analysis*: panels provide valuable data for examining changes over time within the same subjects, enabling detailed trend analysis;
2. *Causal inference*: researchers can better infer causal relationships by tracking the same respondents, as variations can be linked directly to prior states or exposures;
3. *Reduced variability*: since the same individuals are surveyed, differences due to between-subject variability are minimised, leading to more precise estimates;
4. *Enhanced data depth*: panel surveys allow researchers to collect longitudinal data, which provides richer, more nuanced insights than a single point-in-time cross-sectional survey.

Instead, the following issues can be considered drawbacks:

1. *Attrition*: sampled units may drop out over time, leading to a smaller and potentially less representative sample (attrition bias);
2. *Panel conditioning*: repeated questioning may influence respondents' behaviour or attitudes, potentially skewing results;
3. *Complexity in management*: maintaining a consistent panel involves significant logistical challenges, including contact management and retention strategies;
4. *Initial cost and effort*: establishing a panel requires a higher upfront investment in recruitment and baseline data collection;
5. *Risk of non-representativeness*: over time, the panel may become less representative of the broader population due to selective attrition or demographic shifts.

We can observe that many of the above issues do not apply to the LUCAS case. In particular, among the drawbacks, only the last one (risk of non-representativeness) may apply.

By weighing these advantages and disadvantages, researchers can determine whether incorporating a panel component into a periodic survey aligns with their study goals and resources.

The panel component in a LUCAS sample would consist of the Master points already observed in previous rounds of the survey, that are explicitly selected for observation in the current round, based on an *explicit* decision to maintain them in the new sample. It must not be confused with the overlap between two rounds of the survey, which is a random result of the intersection of two consecutive samples.

When a *panel* component is decided, its complement is the *rotation* component.

Three different strategies will be considered in the following:

1. A complete panel approach (panel rate = 1 and rotation rate = 0);
2. A balance between panel and rotation components (panel rate between 0 and 1);
3. A new independent sample selection.

Before illustrating such strategies, let us consider the overlapping rates between previous samples that occurred so far.

Each one of the LUCAS samples, starting from the first round (2009), and ending with the last one (2022), was designed as a cross-sectional sample of the whole frame (the Master Sample).

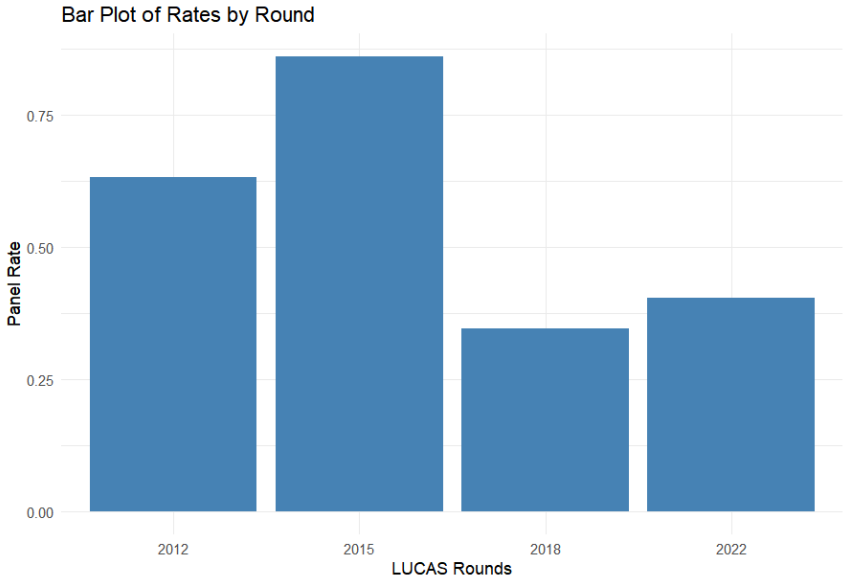
We define the overlapping sample component of each round (hereafter referred to as a wave) as the subset of points that were also observed in the previous round, not necessarily entirely planned as such: when the sampling rate is high, as in the LUCAS case, part of the panel component may result from the casual overlap of the selected points.

Considering the various rounds of LUCAS, the incidence of the overlapping component has been quite erratic (see Figure 1).

The maximum rate was in 2015 (87 %), the minimum in 2018 (34 %); it was 63 % in 2012 and 40 % in 2022. The incidence of the overlapping component was not planned, or, if planned, not based on a rigorous methodology aimed at determining the best rate to ensure the maximization of the accuracy of the estimates.

Figure 1:

Overlapping rates in LUCAS rounds (2012-2022)



In the following, we will perform some simulations assuming as panel the whole overlapping part of the sample. Actually, the overlapping of two samples should be considered a panel component only if such overlapping is planned, in other words, if the selection process in the two samples is not independent. But in this study, we are interested in measuring the total gain in terms of precision that we could have from this overlapping part, due to the covariance component of the total sampling variance calculated for the change estimates. In any case, this is justified by the fact that the non-random component of this overlapping part due to the selection of points in the take-all strata is preponderant.

Two different types of estimates will be considered:

1. The *level* estimates: these concern the totals or means of the parameters of interest at a certain date (for instance, the percentage of artificial land cover in 2022);
2. The *change* estimates, these relate to the variations in the totals or means of the parameters of interest between two different dates (for instance, the increase or decrease in the percentage of artificial land cover from 2018 to 2022).

Let $\mathrm{var}(\overline{Y}_k)$ be the variance of the level estimate for round k, the variance of the change estimate is given by:

$$\mathrm{var}(\overline{Y}_2 - \overline{Y}_1) = \mathrm{var}(\overline{Y}_1) + \mathrm{var}(\overline{Y}_2) - 2\mathrm{cov}(\overline{Y}_{1,}, \overline{Y}_{2,})$$

Looking at the previous formula, the variance in the change estimates is minimized (precision is maximized) when the panel component coincides with the total sample because the covariance component is maximized [1]). On the contrary, as will be shown in the next paragraph, the variance in the level estimates can be minimized by balancing the panel component and its complement, the rotation component.

([1]) That is due that same values are expected between two rounds and therefore the relation is positive and the covariance is greater.

2.1. Methodological approach for the panel rate determination

There is often a trade-off between the precision of the level estimates and the one of the change estimates. A higher panel rate typically improves the precision([2]) for the change estimates but might reduce the precision for the level estimates.

It is possible to use optimization techniques to find the panel rate that maximizes the precision of the level estimates, while the precision of the change estimates is always maximized when the panel rate is equal to 1 (full panel sample). A compromise solution can be found between the two different panel rates.

In the following, we will find the value of the panel rate that minimizes the variance of the level estimates.

To do so, we introduce the *composite estimator* (Fuller and Rao, 2001), which is particularly suitable for our task, as its components are clearly referable to the panel and to the rotation part of the sample.

Given two waves of a sampling survey, at time 1 and at time 2, we introduce a composed estimator to estimate the total or the mean of a target variable Y at time 2:

$$\widehat{\bar{Y}_2^*} = \varphi \cdot \left(\bar{y}_2' + \beta(\bar{y}_1 - \bar{y}_1')\right) + (1-\varphi) \cdot \bar{y}_2'' \quad (1)$$

where:

- $\bar{y}_1$ is the estimator on the complete wave ([3]) (that is, estimates produced by the previous round)
- $\bar{y}_1'$ is the estimator on wave 1 (that is, estimates produced using only the panel observations and the values observed in the previous round)
- $\bar{y}_2'$is the estimator on wave 2 (that is, estimates produced using only the panel observations and the values observed in the current round)
- $\bar{y}_2''$ is the estimator on wave 2 (that is, estimates produced using only the observations belonging to the rotation component and the values observed in the current round)

and

- β is the regression coefficient of the model([4]) fitted using the panel component data and considering as dependent variable the y_2', and as explanatory variable the y_1';
- φ is a parameter used to assign relative importance to the panel component, and to the rotation part.

([2]) It is important to underline that in the following we refer to the *precision* of the estimates, not to their *accuracy*. While the accuracy of an estimate takes into consideration both the components of the Mean Square Error (i.e. bias and variance), the precision depends only on the sampling variance of an estimate (the higher the variance of the estimates, the lower their precision). A higher variance can lead to lower precision, as it indicates greater variability in the estimates. This means that repeated measurements may not consistently yield the same result. On the other hand, accuracy can be affected by both variance and bias.
([3]) For example, there are two components of the estimator a) panel points (2022 - 2018) and b) points only in 2018. Then these two components are aggregated using the weight.
([4]) The regression coefficients are a statistical measure used to describe the average functional relationship between variables. One variable is dependent, while the others are independent, and they measure the degree of dependence of one variable on the other(s).

The value of φ that minimizes the total variance([5]) is:

$$\varphi = \frac{V(\bar{y}_2'')}{V\left(\bar{y}_2' + \beta(\bar{y}_1 - \bar{y}_1')\right) + V(\bar{y}_2'')} \quad (2)$$

As a first approximation (assuming a simple random sample instead of a stratified random sample), the variance of the estimator can be calculated as:

$$V\left(\widehat{\bar{Y}_2^*}\right) = \frac{(1 - \rho^2\,\bar{\lambda})}{1 - \rho^2\,\bar{\lambda}^2}\frac{S_2^2}{n} \quad (3)$$

where:

- ρ is the Pearson coefficient of correlation([6]), calculated using the values of y_2', (the target variable observed in the panel data at wave 2), and y_1' (the target variable observed in the panel data at wave 1);
- S_2^2 is the variance of the target variable in wave 2;
- λ is the panel component and $\bar{\lambda}$ the rotation component ($\lambda = 1 - \bar{\lambda}$).

Using this equation, we can find the value of λ that minimizes the variance $V\left(\widehat{\bar{Y}_2^*}\right)$ of the estimator.

For the estimation of the mean difference between two survey rounds the following estimator can be used:

$$\hat{\Delta} = \widehat{\bar{Y}_2^*} - \widehat{\bar{Y}_1^*} \quad (4)$$

where $\widehat{\bar{Y}_1^*}$ is obtained using the same formula used for $\widehat{\bar{Y}_2^*}$ and changing the indexes 1 and 2: in practice, we estimate the parameter Y at time 1 using also the values observed at time 2, and the parameter Y at time 2 using also the values observed at time 1. It should be noted that $\widehat{\bar{Y}_1^*}$ in equation (4) represents a revised estimate of the estimate produced by the previous round of the survey in light of the current LUCAS 2022 survey results.

2.2. Optimal panel rates

Considering the variables Land Cover and Land Use observed by the LUCAS survey at each round, we derive a binary variable for each of their values.

For instance, considering the LC value 'Artificial,' we have:

$$LC_{art} = \begin{cases} 1 \text{ if } LC = \text{artificial} \\ 0 \text{ otherwise} \end{cases}$$

Using this derived variable LC_{art}, we aim to determine the optimal panel rate, the rate that minimizes the total variance defined in equation (3).

([5]) The two parts of the composite estimator are independent because they are based on two different parts of the same sample. Hence, the covariance between $\widehat{\bar{Y}_2^*}$ and $\bar{y}_2''$ is zero.

([6]) The Pearson coefficient of correlation measures the linear relationship between two variables. It ranges from -1 to 1, where r=1 indicates a perfect positive linear relationship, r=−1 indicates a perfect negative linear relationship, and r=0 indicates no linear relationship.

It is worth noting that equation (3) assumes the same value, $\frac{S_2^2}{n}$, both with λ=0 (no panel component) and with $\lambda = 1$ (the whole sample is a panel). In all other cases ($0< \lambda < 1$) equation (3) assumes a value smaller than 1. Thus, it makes sense to look for an optimal value in the range [0,1]. This optimal value depends on ρ.

In equation (3) the value of n is the sample size of the 2022 round, which is 399,652 points.

The calculation of the variance of $\mathrm{LC}_{\mathrm{art}}$ at time 2 is straightforward, and it is 0.046326.

The calculation of the correlation index for $\mathrm{LC}_{\mathrm{art}}$:

$$\rho_{\mathrm{LC}_{\mathrm{art}}} = \frac{\sum(\mathrm{LC}'_{\mathrm{art}_2} - \overline{\mathrm{LC}}'_{\mathrm{art}_2})(\mathrm{LC}'_{\mathrm{art}_1} - \overline{\mathrm{LC}}'_{\mathrm{art}_1})}{\sqrt{\sum(\mathrm{LC}'_{\mathrm{art}_2} - \overline{\mathrm{LC}}'_{\mathrm{art}_2})^2 \sum(\mathrm{LC}'_{\mathrm{art}_1} - \overline{\mathrm{LC}}'_{\mathrm{art}_1})^2}}$$

is performed on the panel set of data (points in common for 2018 and 2022 waves), and it results in 0.8362.

Finally, to show the effect on the variance of different values of the panel rate, and to determine the best one, we simply iterate the value of λ from 0 to 1 with step=0.01, and obtain the result shown in Figure 2. The red circle indicates the point of minimum, which in this case is placed at the x-axis=0.35.

The same procedure has been applied to all the other target estimates (derived from both Land Cover and Land Use), and the relative plots are contained in Annex 5 – Panel rates determination. The results are reported in Table 1 and Table 2.

Figure 2:

Determination of the optimal rotation rate for LC 'Artificial' estimate

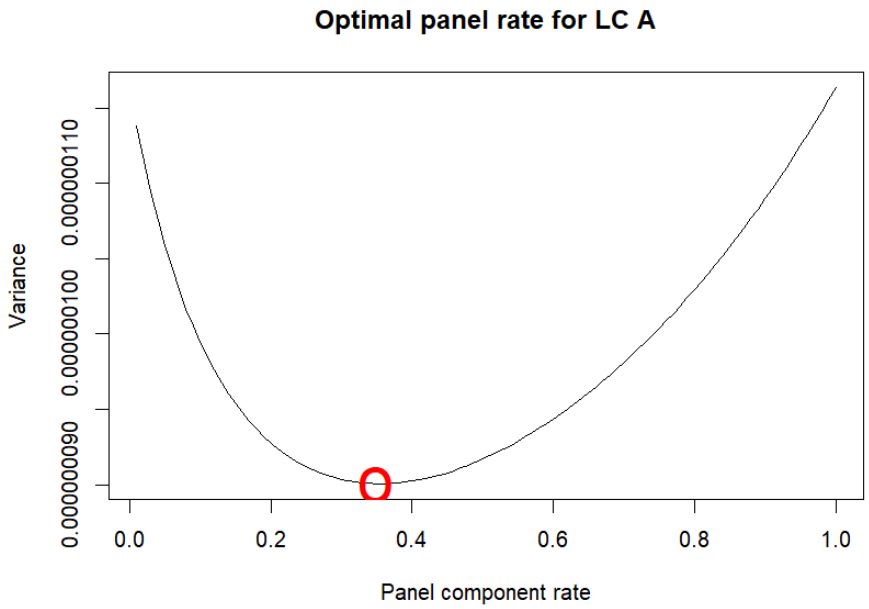


Table 1:

Optimal panel rates for Land Cover estimates

LAND COVER	TOTAL VARIANCE	RHO	PANEL RATE
A - Artificial	0.000000096	0.84	0.35
B - Cropland	0.000000393	0.77	0.39
C - Woodland	0.000000410	0.88	0.32
D - Shrubland	0.000000152	0.71	0.41
E - Grassland	0.000000403	0.74	0.40
F - Bareland	0.000000094	0.36	0.48
G - Water	0.000000083	0.89	0.32
H - Wetland	0.000000040	0.86	0.34

Table 2:

Optimal panel rates for Land Use estimates

LAND USE	TOTAL VARIANCE	RHO	PANEL RATE
U1 – Primary Sector	0.000000442	0.86	0.34
U2 – Secondary Sector	0.000000009	0.77	0.39
U3 – Tertiary Sector	0.000000208	0.87	0.33
U4 – Abandoned/unused	0.000000325	0.82	0.36

It is interesting to note that, in the case of level estimates, there is an inverse relation between the values of the coefficient of correlation *rho*, and the optimal values of the optimal panel rate, as is evident in Figure 3 and Figure 4.

For instance, LC estimate 'G' (Water) is characterized by a high rho value (89 %) and the minimum panel rate (32 %), while, in the case of LC estimate 'F' (Bareland), we have the minimum value of rho (36 %) and the maximum value of optimal panel rate (48 %).

Figure 3:

Optimal panel rates vs coefficients of correlation for Land Cover estimates

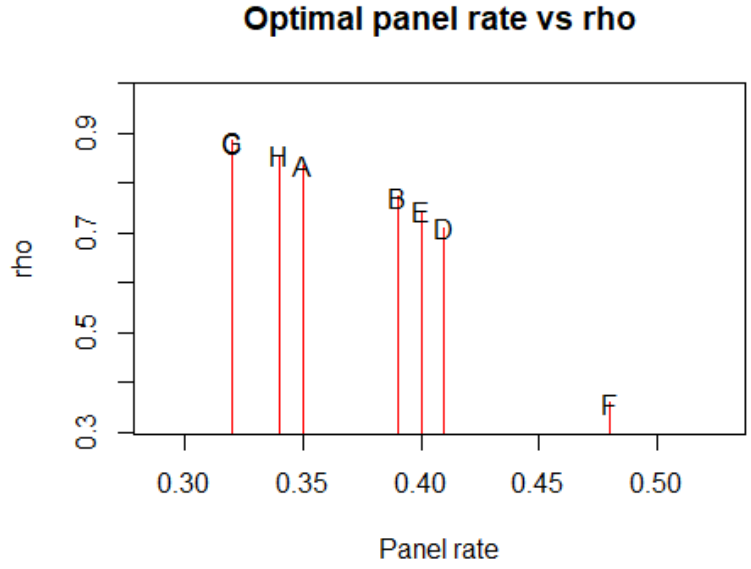
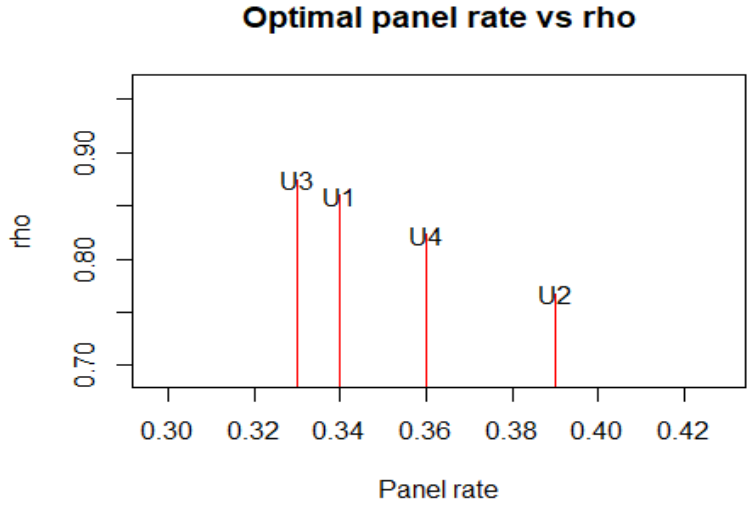


Figure 4:

Optimal rotation rates vs coefficients of correlation for Land Use estimates



If we consider only the level estimates, we should choose a panel component rate in the interval 32%-48%. If we consider LC “F” as an “outlier” the range could be 32%-41%.

Instead, if we consider the change estimates, it can be proved that the maximum precision can be obtained using a panel rate equal to 1 (the same units in both waves).

Since the two objectives are conflicting, the practical solution could be a compromise between the optimal panel rate determined for the level estimates, and the maximum value of 100 % of the panel component that maximizes the precision of the change estimates. A practical solution has to be further discussed if it is worth maintaining the panel component from the last two or three surveys.

3

Level estimates: the composed estimator

In the previous section, we have introduced the composed estimator, particularly suitable for the determination of the panel/rotation component rates. It is advisable to ensure that this estimator is not less efficient than the one currently used for the calculation of LUCAS estimates: if not, the indications received in terms of best panel/rotation rates would be less acceptable. Once a planned panel component is introduced in the LUCAS survey, the composed estimator could substitute the usual calibration estimator currently employed for the production of the estimates.

For these reasons, the composed estimator has been applied to the 2022 LUCAS data, and the resulting estimates and related precision compared to those obtained by the current estimator. The current estimator is a calibrated one and it is fully described in Ballin et al (2023). At the EU level, the results of the comparison are reported in Table 3 in Figures 5-6.

Table 3:

Estimates of the fraction of LC and LU area and related coefficients of variation obtained by the composed estimator and the current estimator

Land Cover/Land Use	Estimates (composed)	CVs (composed)	Estimates (current)	CVs (current)
A - Artificial	0.044	0.006	0.039	0.010
B - Cropland	0.226	0.002	0.217	0.003
C - Woodland	0.409	0.002	0.383	0.002
D - Shrubland	0.061	0.008	0.070	0.009
E - Grassland	0.172	0.004	0.178	0.005
F - Bareland	0.041	0.009	0.047	0.010
G - Water	0.031	0.005	0.049	0.004
H - Wetland	0.016	0.009	0.016	0.016
U1 – Primary Sector	0.783	0.001	0.713	0.002
U2 – Secondary Sector	0.002	0.043	0.003	0.034
U3 – Tertiary Sector	0.089	0.005	0.099	0.006
U4 – Abandoned/unused	0.126	0.006	0.185	0.006

Figure 5:

Comparison between estimates obtained by the composed and the current estimators

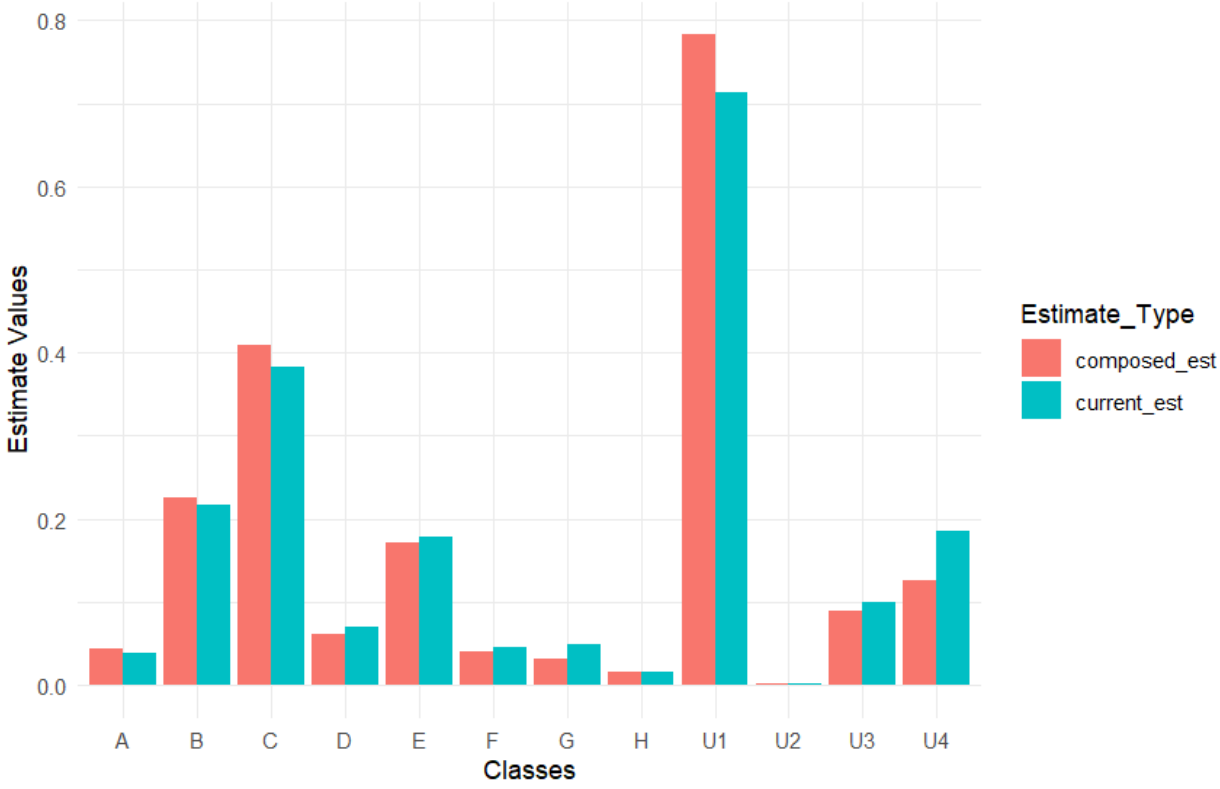
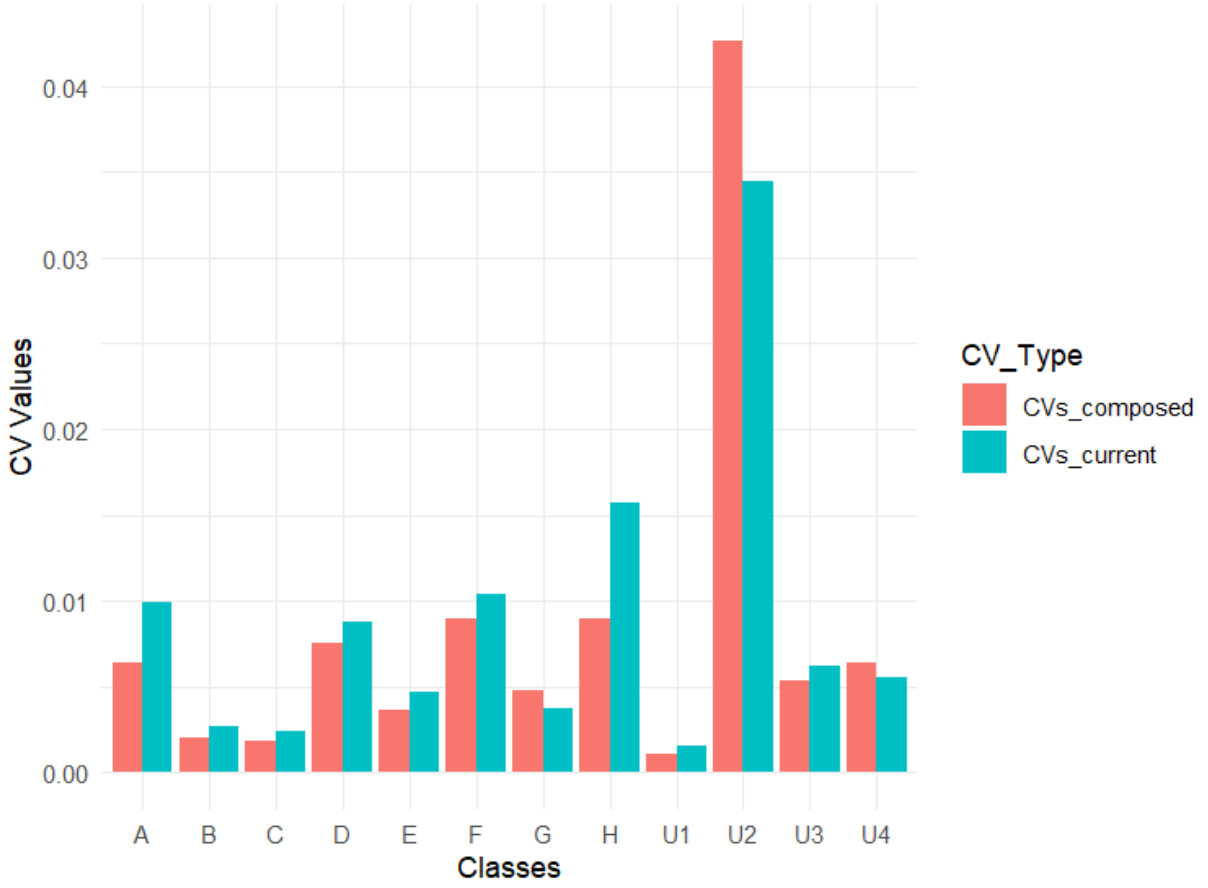


Figure 6:

Comparison between Coefficients of Variation obtained by the composed and the current estimators



Some considerations:

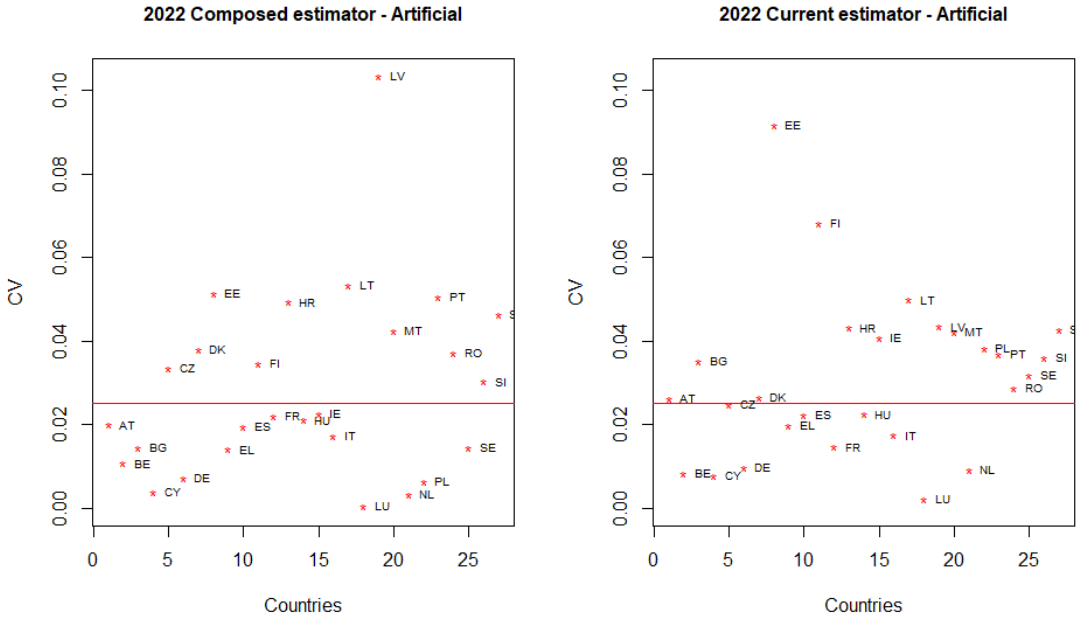
- Differences in estimates are in some cases non-negligible, particularly for Land Use classes;
- Coefficients of variations obtained by the composed estimator are in general lower than those obtained by the current estimator in the case of the Land Cover Classes, not for the Land Use classes.

The above is for the EU level.

At the country level (NUTS0), we have compared the coefficients of variation related to the same Land Cover or Land Use class, related to the estimates produced in the 27 EU countries by the two estimators. In Figure 7 is reported the particular case related to the Land Cover class 'Artificial'. In this graph, we can identify the values of the coefficients of variation for each single country.

Figure 7:

Comparison between Coefficients of Variation obtained by composed and current estimators in the different EU countries for Land Cover class 'Artificial'



To permit an overall evaluation of the different precision levels obtained by the two estimators, in Table 4 we report the mean, min, and max values of the coefficients of variation in 2018-2022.

Table 4:

Mean, min and max values of the coefficients of variations in 2018 and 2022

LC and LU classes	CVs of composed estimator			CVs of current estimator		
	Mean	Min	Max	Mean	Min	Max
A	0.02847	0.00051	0.10337	0.03112	0.00211	0.09157
B	0.01326	0.00050	0.05388	0.01256	0.00000	0.04001
C	0.01055	0.00063	0.03662	0.01043	0.00094	0.03401
D	0.02678	0.00216	0.08745	0.04569	0.00286	0.16693
E	0.01695	0.00071	0.06547	0.01894	0.00072	0.06977
F	0.02552	0.00051	0.09580	0.05223	0.00000	0.14995
G	0.06092	0.00095	0.73926	0.01623	0.00000	0.03757
H	0.03767	0.00000	0.16457	0.04527	0.00000	0.20468
U1	0.00617	0.00033	0.01659	0.00490	0.00032	0.01152
U2	0.18553	0.00197	0.72972	0.15164	0.00000	0.46277
U3	0.03635	0.00230	0.21302	0.03143	0.00173	0.14730
U4	0.04092	0.00000	0.12958	0.03516	0.00221	0.15660

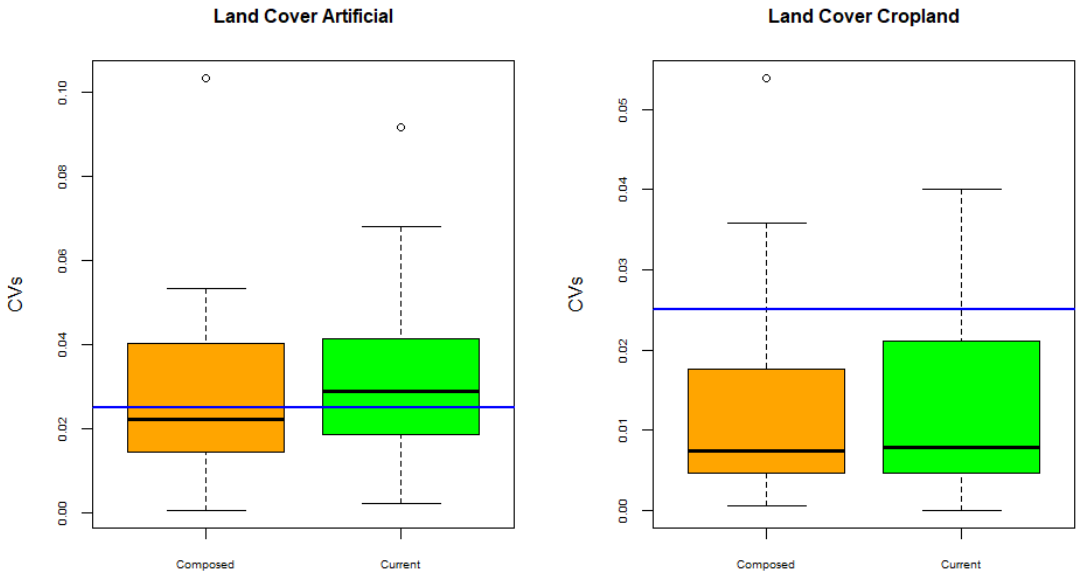
In each plot of Figure 8, the boxplots of the coefficients of variation related to the same Land Cover or Land Use class, concerning the estimates produced in the 27 EU countries by the two estimators, are compared.

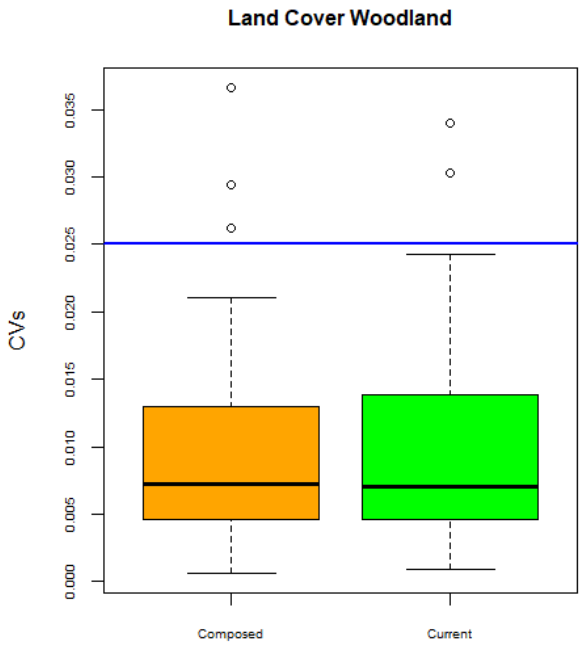
Analysing the values in Table 4 and the boxplots in Figure 8, there is no clear evidence of one estimator being superior to the other in terms of precision.

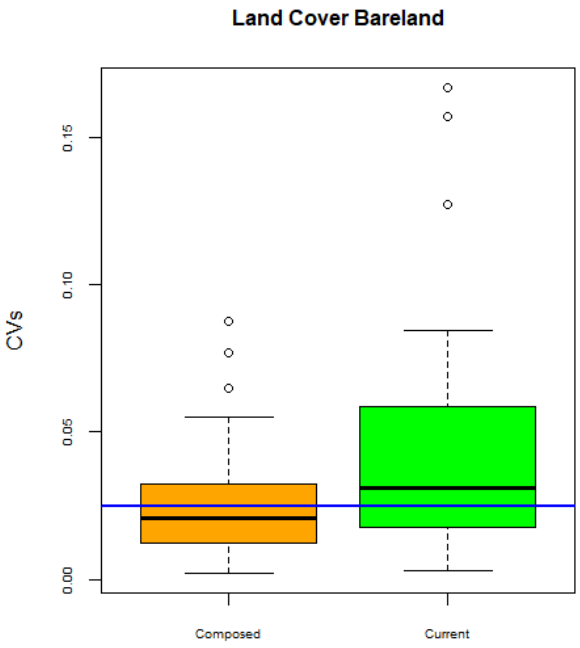
For instance, considering the Land Cover classes, the composed estimator seems to be convenient, as its coefficients of variation are lower (the precision higher – blue horizontal line) in 'Artificial,' 'Bareland,' 'Grassland,' and 'Shrubland,' while in the other cases ('Cropland,' 'Woodland,' 'Water,' and 'Wetland,') they are more or less equivalent. Instead, for the Land Use classes, the current estimator seems to be more efficient.

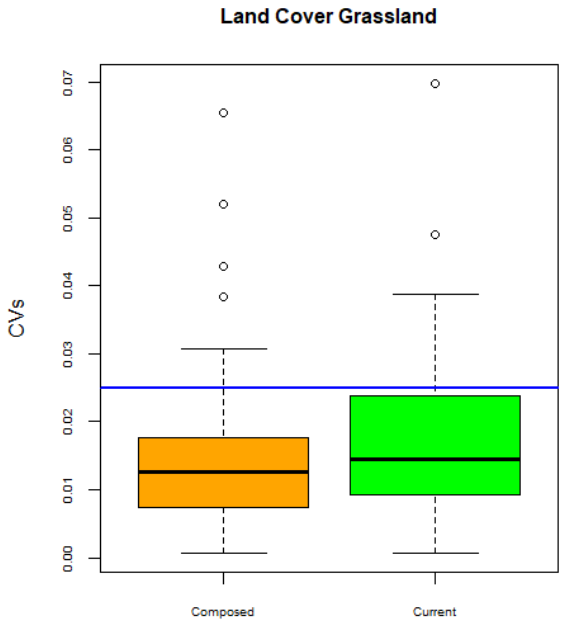
Figure 8:

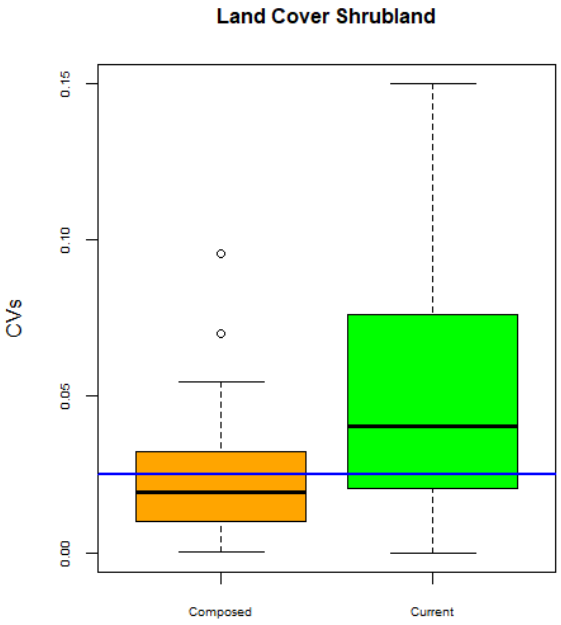
Boxplots of the coefficients of variation obtained by composed and current estimators in the different EU countries

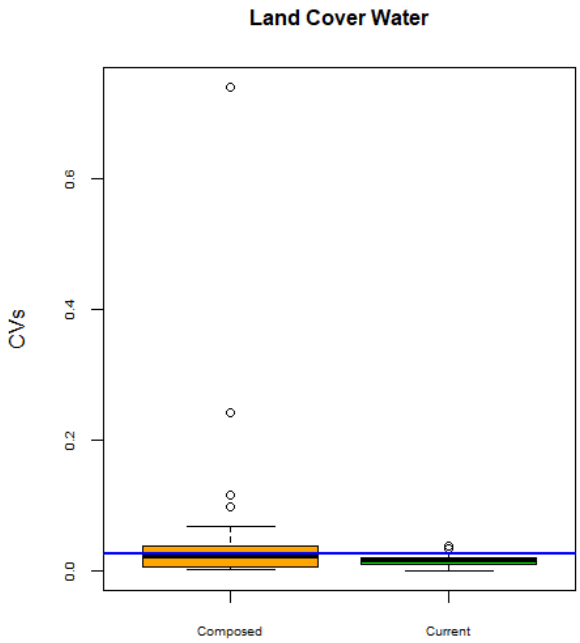


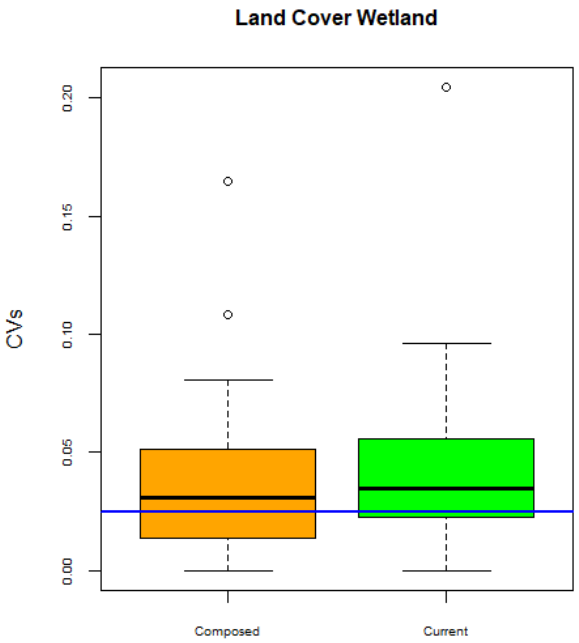

Land Cover Woodland
CVs
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
Composed
Current

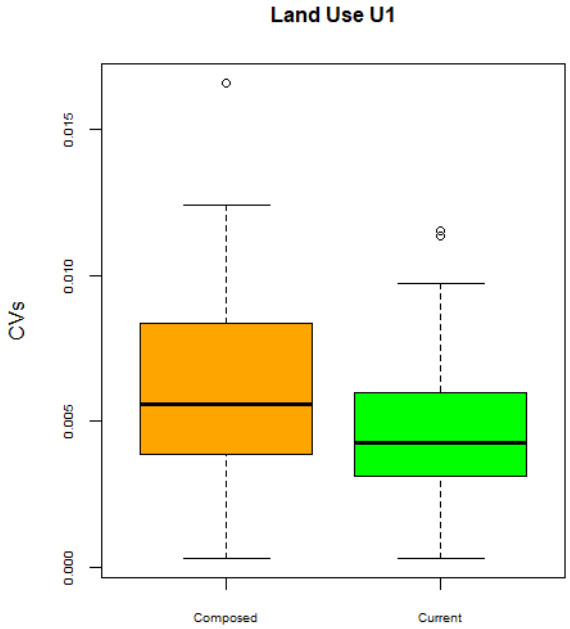

Land Cover Bareland
CVs
0.00
0.05
0.10
0.15
Composed
Current

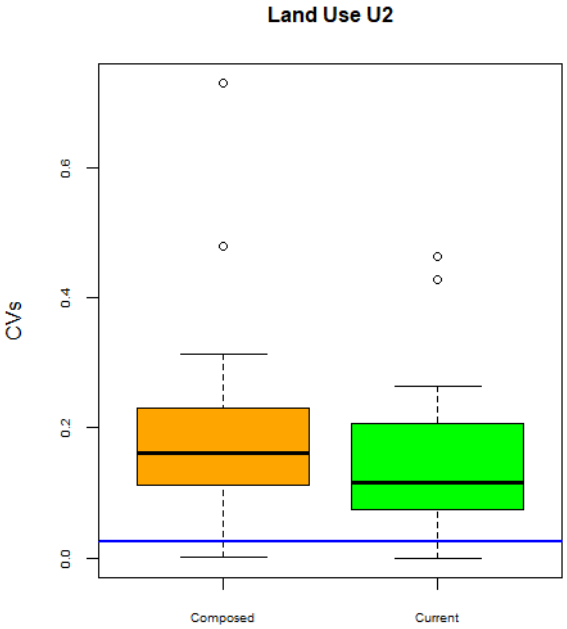

Land Cover Grassland
CVs
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Composed
Current

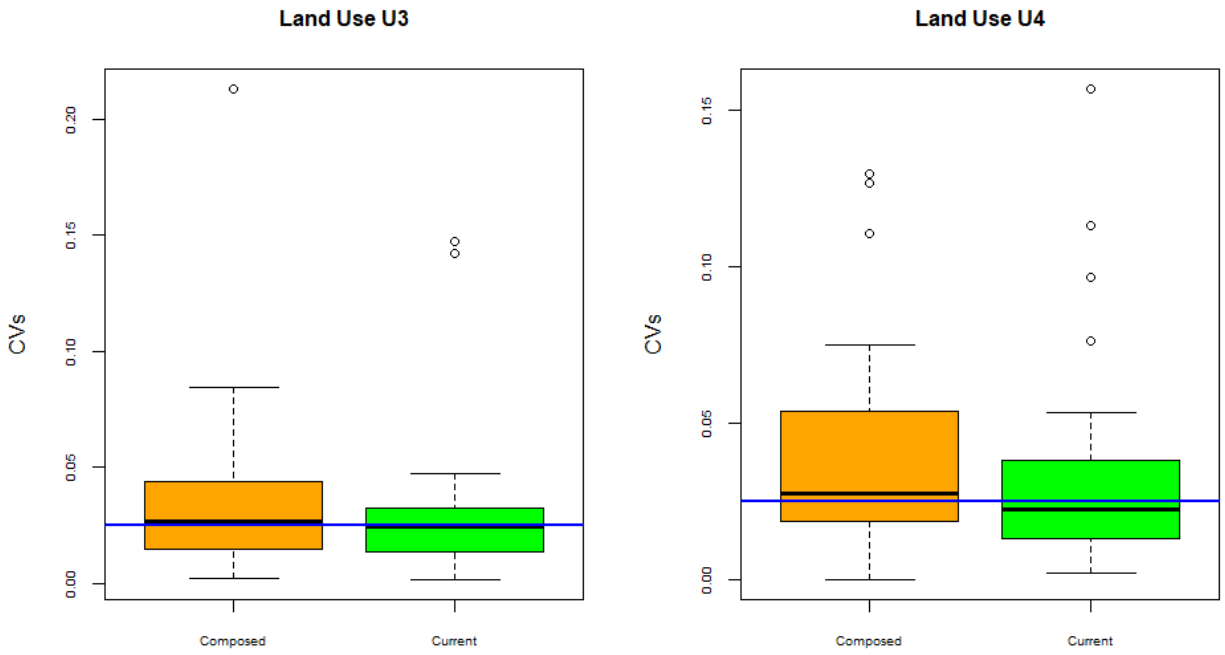

Land Cover Shrubland
CVs
0.00
0.05
0.10
0.15
Composed
Current


Land Cover Water
CVs
0.0
0.2
0.4
0.6
Composed
Current


Land Cover Wetland
CVs
0.00
0.05
0.10
0.15
0.20
Composed
Current


Land Use U1
CVs
0.000
0.005
0.010
0.015
Composed
Current


Land Use U2
CVs
0.0
0.2
0.4
0.6
Composed
Current


Land Use U3
CVs
0.00
0.05
0.10
0.15
0.20
Composed
Current
Land Use U4
CVs
0.00
0.05
0.10
0.15
Composed
Current

4 Accuracy of change estimates

The importance of the accuracy of the change estimates was particularly evident in the 2018 round, when the estimates of the variations in artificial areas with respect to 2015 were negative in many regions, and in some of them, this difference appeared to be significant.

The calculation of the variations was carried out in two steps:

1. First, the point estimates were calculated for both 2015 and 2018;
2. Then, their difference was calculated.

The significance of the difference was tested considering the confidence intervals of both estimates: in case of overlap, their difference was deemed as not significant, otherwise it was significant.

What is the situation with regard to the 2022 vs 2018?

We analysed this new situation using 'svyDelta', a new function of the R package ReGenesees (Zardetto, 2015), which takes into account the composition of the sample (panel and rotation components), and produces the estimates in a more efficient way, while also correctly calculating their sampling variance([7]).

We used the *svyDelta* function to calculate the differences in the estimates of Artificial area between 2018 and 2022, considering EU level, and by NUTS0 (countries) and NUTS2 (regions).

A key factor in computing correctly the sampling variance of the differences in the estimates is in considering the two samples (2018 and 2022) as not independent, otherwise, the covariance terms would not be considered.

([7])From the documentation of the ReGenesees package:
'Function svyDelta computes estimates and sampling errors of a Measure of Change from two not necessarily independent samples. For 'Measure of Change' we mean here any function that can be used to compare cross-sectional estimates of two population parameters derived from two samples. The easiest Measure of Change is the simple difference between the estimators of the same population parameter referred to two distinct survey occasions. When the two samples being analysed are independent, estimating the sampling variance of a simple Measure of Change is fairly straightforward: one just has to add variance estimates derived from the two samples. The problem becomes non-trivial when the two samples are not independent, because sampling covariance terms come into play. If the samples in design1 and design2 are not independent, function svyDelta estimates the needed sampling covariance terms following the method of Berger and Priam (Berger and Priam, 2016). Note that this method derives design-based covariance estimates from estimates of correlation (rho), which are computed in the first place.'

We know that the two samples were formally selected independently (except the small Soil module panel component). However, we must consider the fact that a completely independent selection would result in a panel component of only about 13 % (resulting from the product of the two sampling rates, 34 % in 2018 and 40 % in 2022), whereas we observe a much higher panel component, about 40 %. This depends on the sampling designs of the two rounds, which were based on many strata that were censused: the number of points included in the panel component that was selected with probability equal to 1 in 2018 and in 2022 represents 70 % of the panel, which is the quota exceeding the 13 % that would be expected in the case of independence. For this reason, we cannot consider the two samples as independent, and therefore we can take into consideration the covariance component in computing the total sampling variance of the change estimates. To make the two estimates (2018 and 2022) comparable, data have been calibrated by areas (calibration models could be more complex).

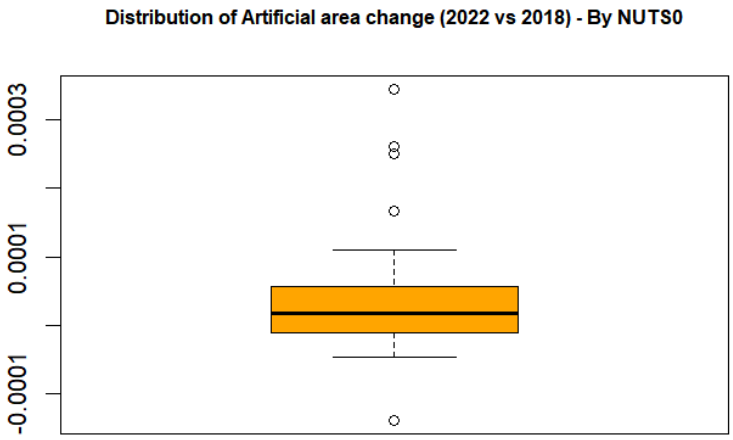
At the EU level, the increase in artificial area is equal to +0.0012, with its 95 % confidence interval being 0.0003-0.0021. Equivalently, the coefficient of variation is equal to 36 % (it is worth noting that the high coefficient of variation is due to the very small value of the estimate).

Considering the 27 countries defined by NUTS0 classification, Figure 9 shows the distribution of the change estimates. More specifically, this is the situation for the 27 countries:

- In 10 out of 27 countries, the difference is negative;
- But for all of the 10 negative differences, the corresponding confidence intervals include the zero value, so we can conclude that this negative difference is not significant.

Figure 9:

Distribution of the change estimates (2018-2022) by NUTS0 for 'Artificial' area change

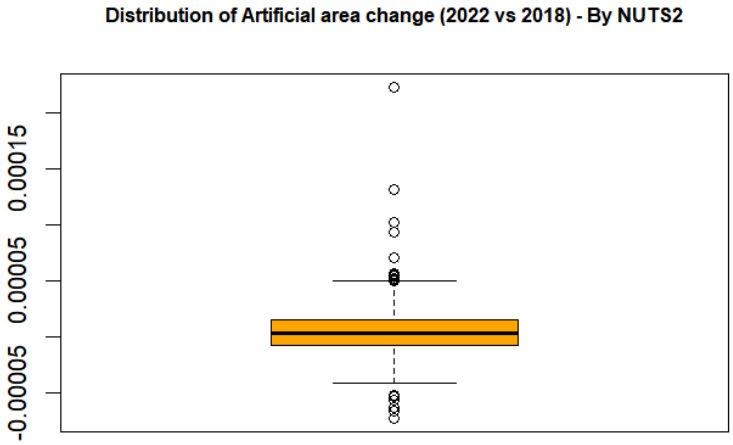


This is the situation for the 230 regions: in 96 regions out of 230, the difference is negative. But for all of the 96 negative point estimates, the corresponding confidence intervals include 0, so we can conclude that this negative difference is not significant.

The complete distribution of the 230 change estimates is reported in Figure 10.

Figure 10:

Distribution of the change estimates (2018-2022) by NUTS2 for 'Artificial' area change



Therefore, it seems that the problem that occurred in 2018 is no longer evident in 2022. This positive outcome could depend on the increased rate of the panel component (34 % in 2018 and 40 % in 2022).

5 Conclusions

In the previous paragraphs, some technical elements useful to understand the consequences of the panel strategies on the accuracy of the estimates have been given.

Turning back to the three strategies:

1. A complete panel approach;
2. A balance between panel and rotation components;
3. A new independent sample selection,

an absolute best solution cannot be indicated because it depends on:

- The importance given to each statistical objective (levels vs changes);
- The observed accuracies (satisfactory or not);
- The module requirements (relevance given to observations on the same points);
- The management of surveyors' activities (knowledge of accessibility of each point, changes FI/PI for some points, etc.).

Each of these aspects pushes in different directions.

A complete panel approach (panel rate equal to 1)

This strategy implies carrying out the next round of LUCAS on the same units used for the 2022 survey (100 % panel). It is the simplest strategy to implement, because it does not require a new sample design or a new sample selection process.

This is also the best strategy to avoid or to limit the presence of anomalous changes, for instance, decreases in artificial areas or big changes in water areas.

It allows the analysis of changes not only at the macro level, but at the micro level too([8]), opening the door to the development of statistical models to predict changes.

It has some advantages from the organizational point of view too:

- Surveyors are familiar with points and the path to reach them;
- Some points can be switched from 'observed in the field' to 'photo-interpreted' based on metadata collected in the 2022 survey.

On the other side, this strategy, excluding refinements obtainable with more sophisticated estimators, could not improve the accuracies obtained with the last round of the survey, because the design of the sample is given and no better allocation would be permitted.

Therefore, in case the current accuracy of the level estimates is not considered completely satisfactory, the full panel would not be advisable.

Balance between panel and rotation components (panel rate between 0 and 1)

The technicalities of this approach have been illustrated with reference to the composed estimator.

Summarizing the results of the study the following pros and cons of an intermediate panel rate are:

- replacing some units of the sample with some new fresh units allows the adoption of composed estimators. Such an estimator improves the accuracy of the level estimates.
- the rotation to be adopted could be a compromise between the optimal rate for each LC in each country and organizational considerations.

The estimates of the changes will not be optimal because it is not a strict panel.

The sample selection of the new units is not an easy task without losing control of the sample randomness. Such difficulties are mainly due to some constraints in the management of the modules and to the limited size of some strata which contain few points.

A new independent sample selection (panel rate equal to 0)

Since the current sample design and the new one could be based on different stratifications, the management of the panel component (maintaining the control of the randomness) could be a hard task.

If the overlap with the current sample is not managed (independence of the samples), we would obtain optimality in the level estimates, but not gain any efficiency in estimates of changes.

The assignment of the observation type (in the field or photo-interpreted) would be based mainly on predicted values and not on observed characteristics.

[8] In this context, macro-level analysis concerns the comparison of estimates referred to the same domain in different reference times (such estimates can be obtained by independent samples), while micro-level analysis is referred to the search of "models" linking the changes and the characteristics of the points-

Final considerations

Introducing a planned sample is convenient because of the LUCAS survey's specific characteristics and because it would increase the precision of the change estimates.

However, indicating a precise rate of such a panel component is difficult. Looking at the level estimates of the various LC and LU classes, it should range from 32% to 48% (or from 32% to 41% if we consider the F class as an outlier). Considering the change estimates, it should be 100%.

Fixing too high levels of the panel component, though, could introduce too tight constraints which could cause difficulties, for example, for the module sub-samples selection in the next LUCAS round. Or, if it will be necessary to further optimize the sample design to increase the precision of some estimates in given sub-domains.

In conclusion, the determination of a precise panel component rate will be only possible when all the requirements of the next LUCAS survey and of the modules will be clear.

6

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Introduction of a panel component in the LUCAS survey and the use of a composed estimator for statistics production

The paper presents a statistical study aimed at determining the optimal rate of panel sample points for the European Land Cover and Land Use Survey (LUCAS) survey. Furthermore, the pros and cons of adopting a composed estimator for statistics production by taking full advantage of a panel component are investigated.

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